```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
dataset = pd.read_csv('/content/drive/MyDrive/robotic_arm_positions_normal02.csv')
X = dataset.iloc[:, 4:7].values
y = dataset.iloc[:, 0:4].values
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.20, random_state = 0)
```

```
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeClassifier, ExtraTreeClassifier, ExtraTreeRegressor
from sklearn.ensemble import RandomForestClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.svm import SVR
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.ensemble import AdaBoostRegressor, BaggingRegressor

regressor = BaggingRegressor(ExtraTreeRegressor(random_state=0), random_state=0, n_estimators=100)
#regressor = RandomForestClassifier(n_estimators=30)

regressor.fit(X_train, y_train)
```

```
BaggingRegressor
estimator: ExtraTreeRegressor
ExtraTreeRegressor
```

```
y_pred=regressor.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error
mean_squared_error(y_test, y_pred)
```

```
5.2043729836606465
```

```
from sklearn.metrics import r2_score
r2_score(y_test, y_pred, multioutput='raw_values')
```

```
array([0.99974182, 0.9965251 , 0.96183694, 0.22517425])
```

```
regressor.predict(X_test)
```

```
array([[ 1.70200000e+02,  2.10341667e+01, -5.72891667e+01,
        -8.74791667e+01],
       [ 1.24800000e+02,  5.05000000e+00, -4.55000000e+01,
        -8.17216667e+01],
       [ 1.64900000e+02,  6.45000000e+01, -5.90000000e+01,
        -8.32641071e+01],
       ...,
       [ 1.74950000e+02,  1.08315238e+02, -7.83152381e+01,
        -8.54297500e+01],
       [ 5.97500000e+01,  9.44363095e+01, -8.90863095e+01,
        -8.59077619e+01],
       [ 5.00000000e-02,  9.95472619e+01, -7.34972619e+01,
        -8.37233333e+01]])
```

```
print(X[0])
print(y[0])
```

```
regressor.predict([[0.35,0,-0.20]])
```

```
array([[  0.    ,  10.25  , -45.2   , -83.1285]])
```

```
import pickle
with open('invkine02.pkl', 'wb') as f:
    pickle.dump(regressor, f)
```

```
with open('invkine02.pkl', 'rb') as f:
    clf = pickle.load(f)
```

```
clf.predict([[0.35,0,-0.20]])
```

```
array([[  0.    ,  10.25  , -45.2   , -83.1285]])
```